

Homework 2

Due Date: End of Week 3, Sunday at midnight

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Q1: A perfectly competitive industry consists of many identical firms, each with a long run average cost function of:

$$TC = 125.Q - 30.Q^2 + 3Q^3$$

What is the equation for the average cost?

What is the equation for the marginal cost?

In the long run how much will each firm produce (in the long run all perfectly competitive identical firms will produce at minimum average cost)?

What is the long run equilibrium price (In the long run $P = \min AC$, i.e., zero economic profits)?

The industry demand curve is:

$$Q_D = 7800 - 12.P$$

How many firms are in the industry?

Q2: Consider the cost minimization problem

$$\text{Min } C = rK + wL \text{ subject to } \bar{Y} = f(K, L)$$

Where C is cost, r is rental on capital, K is Capital, w is the wage, L is labor and Y is income.

Show that, at the optimum, λ^* is equal to the marginal cost wrt to income (Hint: chain rule).

Can you extend the derivation above to show that in equilibrium: $\frac{MPK}{MPL} = \frac{r}{w}$?

Q3: The following is a Stone Geary or LES consumer utility function:

$$u = (x_1 - c_1)^a \cdot (x_2 - c_2)^{1-a}$$

Where: $0 < a < 1$

Where c_1 and c_2 are interpreted as minimum amounts of the good required for subsistence.

Derive the demand functions, discuss their properties, and compare them with the demand functions derived in the Cobb Douglas example.

Hint: Use $y = p_1(x_1 - c_1) + p_1c_1 + p_2(x_2 - c_2) + p_2c_2$ for the constraint.